

```
import pandas as pd
df = pd.read_csv("Titanic-Dataset.csv")
```

```
df.describe()
```

	PassengerId	Survived	Pclass	Age	SibSp \
count	891.000000	891.000000	891.000000	714.000000	891.000000
mean	446.000000	0.383838	2.308642	29.699118	0.523008
std	257.353842	0.486592	0.836071	14.526497	1.102743
min	1.000000	0.000000	1.000000	0.420000	0.000000
25%	223.500000	0.000000	2.000000	20.125000	0.000000
50%	446.000000	0.000000	3.000000	28.000000	0.000000
75%	668.500000	1.000000	3.000000	38.000000	1.000000
max	891.000000	1.000000	3.000000	80.000000	8.000000

	Parch	Fare
count	891.000000	891.000000
mean	0.381594	32.204208
std	0.806057	49.693429
min	0.000000	0.000000
25%	0.000000	7.910400
50%	0.000000	14.454200
75%	0.000000	31.000000
max	6.000000	512.329200

```
df.head()
```

	PassengerId	Survived	Pclass	\
0	1	0	3	
1	2	1	1	
2	3	1	3	
3	4	1	1	
4	5	0	3	

	Name	Sex	Age
SibSp \			
0	Braund, Mr. Owen Harris	male	22.0
1			
1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0
1			
2	Heikkinen, Miss. Laina	female	26.0
0			
3	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0
1			
4	Allen, Mr. William Henry	male	35.0
0			

	Parch	Ticket	Fare	Cabin	Embarked
0	0	A/5 21171	7.2500	NaN	S
1	0	PC 17599	71.2833	C85	C

2	0	STON/O2.	3101282	7.9250	NaN	S
3	0		113803	53.1000	C123	S
4	0		373450	8.0500	NaN	S

```
df.tail()
```

	PassengerId	Survived	Pclass	Name \
886	887	0	2	Montvila, Rev. Juozas
887	888	1	1	Graham, Miss. Margaret Edith
888	889	0	3	Johnston, Miss. Catherine Helen "Carrie"
889	890	1	1	Behr, Mr. Karl Howell
890	891	0	3	Dooley, Mr. Patrick

	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
886	male	27.0	0	0	211536	13.00	NaN	S
887	female	19.0	0	0	112053	30.00	B42	S
888	female	NaN	1	2	W./C. 6607	23.45	NaN	S
889	male	26.0	0	0	111369	30.00	C148	C
890	male	32.0	0	0	370376	7.75	NaN	Q

```
df.shape
```

```
(891, 12)
```

```
df.columns
```

```
Index(['PassengerId', 'Survived', 'Pclass', 'Name', 'Sex', 'Age', 'SibSp', 'Parch', 'Ticket', 'Fare', 'Cabin', 'Embarked'], dtype='object')
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
#   Column      Non-Null Count  Dtype
---  -
0   PassengerId  891 non-null    int64
1   Survived     891 non-null    int64
2   Pclass       891 non-null    int64
3   Name         891 non-null    object
4   Sex          891 non-null    object
5   Age         714 non-null    float64
6   SibSp        891 non-null    int64
```

```
7   Parch      891 non-null   int64
8   Ticket     891 non-null   object
9   Fare       891 non-null   float64
10  Cabin      204 non-null   object
11  Embarked   889 non-null   object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
```

```
df.dtypes
```

```
PassengerId    int64
Survived        int64
Pclass          int64
Name            object
Sex             object
Age            float64
SibSp           int64
Parch           int64
Ticket          object
Fare            float64
Cabin           object
Embarked        object
dtype: object
```

```
df['Survived']
```

```
0      0
1      1
2      1
3      1
4      0
..
886    0
887    1
888    0
889    1
890    0
Name: Survived, Length: 891, dtype: int64
```

```
df.isnull().sum()
```

```
PassengerId    0
Survived        0
Pclass          0
Name            0
Sex             0
Age            177
SibSp           0
Parch           0
Ticket          0
Fare            0
```

```
Cabin          687
Embarked       2
dtype: int64
```

```
df['Survived'].value_counts()
```

```
Survived
0      549
1      342
Name: count, dtype: int64
```

```
!pip install matplotlib
```

```
Collecting matplotlib
```

```
  Downloading matplotlib-3.10.3-cp313-cp313-win_amd64.whl.metadata (11 kB)
```

```
Collecting contourpy>=1.0.1 (from matplotlib)
```

```
  Downloading contourpy-1.3.2-cp313-cp313-win_amd64.whl.metadata (5.5 kB)
```

```
Collecting cycler>=0.10 (from matplotlib)
```

```
  Downloading cycler-0.12.1-py3-none-any.whl.metadata (3.8 kB)
```

```
Collecting fonttools>=4.22.0 (from matplotlib)
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```
  Downloading fonttools-4.59.0-cp313-cp313-win_amd64.whl.metadata (110 kB)
```

```
Collecting kiwisolver>=1.3.1 (from matplotlib)
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  Downloading kiwisolver-1.4.8-cp313-cp313-win_amd64.whl.metadata (6.3 kB)
```

```
Requirement already satisfied: numpy>=1.23 in d:\college\machine learning (ml)\.venv\lib\site-packages (from matplotlib) (2.3.1)
```

```
Requirement already satisfied: packaging>=20.0 in d:\college\machine learning (ml)\.venv\lib\site-packages (from matplotlib) (25.0)
```

```
Collecting pillow>=8 (from matplotlib)
```

```
  Downloading pillow-11.3.0-cp313-cp313-win_amd64.whl.metadata (9.2 kB)
```

```
Collecting pyparsing>=2.3.1 (from matplotlib)
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  Downloading pyparsing-3.2.3-py3-none-any.whl.metadata (5.0 kB)
```

```
Requirement already satisfied: python-dateutil>=2.7 in d:\college\machine learning (ml)\.venv\lib\site-packages (from matplotlib) (2.9.0.post0)
```

```
Requirement already satisfied: six>=1.5 in d:\college\machine learning (ml)\.venv\lib\site-packages (from python-dateutil>=2.7->matplotlib) (1.17.0)
```

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Downloading matplotlib-3.10.3-cp313-cp313-win_amd64.whl (8.1 MB)
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Downloading contourpy-1.3.2-cp313-cp313-win_amd64.whl (223 kB)
Downloading cycler-0.12.1-py3-none-any.whl (8.3 kB)
Downloading fonttools-4.59.0-cp313-cp313-win_amd64.whl (2.2 MB)
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Downloading kiwisolver-1.4.8-cp313-cp313-win_amd64.whl (71 kB)
Downloading pillow-11.3.0-cp313-cp313-win_amd64.whl (7.0 MB)
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```

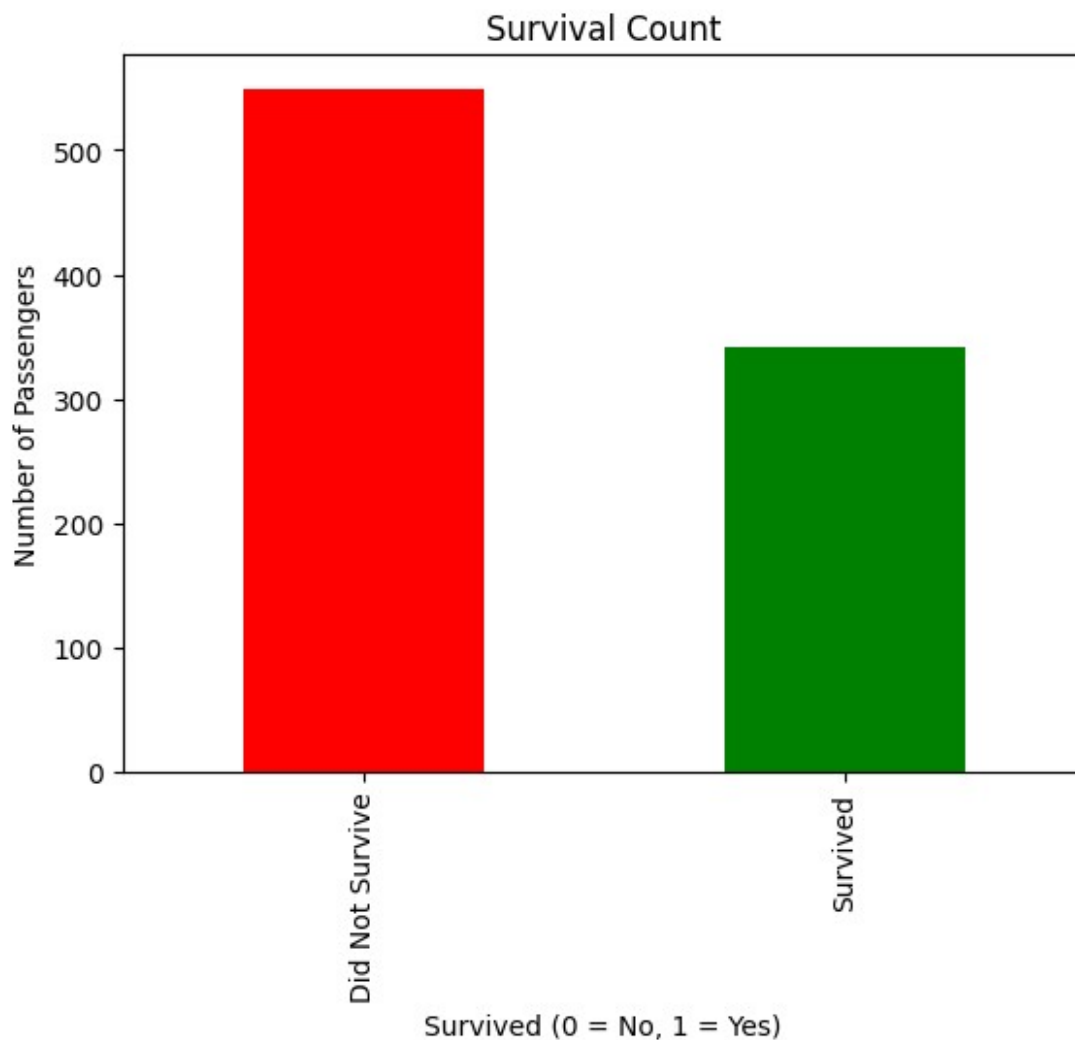

[illegible]

[illegible]

Successfully installed contourpy-1.3.2 cycler-0.12.1 fonttools-4.59.0
kiwisolver-1.4.8 matplotlib-3.10.3 pillow-11.3.0 pyparsing-3.2.3

```
import matplotlib.pyplot as plt
df['Survived'].value_counts().plot(kind='bar', color=['red', 'green'])
plt.title('Survival Count')
plt.xlabel('Survived (0 = No, 1 = Yes)')
plt.ylabel('Number of Passengers')
plt.xticks([0, 1], ['Did Not Survive', 'Survived'])
plt.show()
```

Matplotlib is building the font cache; this may take a moment.



```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("Titanic-Dataset.csv")
```

```
# Create a grouped dataframe: count of survivors by 'Sex' and 'Pclass'
grouped = df.groupby(['Pclass', 'Sex'])['Survived'].sum().unstack()

# Plot the stacked bar chart
grouped.plot(kind='bar', stacked=True, color=['skyblue', 'lightpink'],
figsize=(8, 6))

plt.title('Survival Count by Class and Gender')
plt.xlabel('Passenger Class')
plt.ylabel('Number of Survivors')
plt.legend(title='Gender')
plt.xticks(rotation=0)
plt.grid(axis='y')
plt.show()
```

